

Introduction

Rapid advances in artificial intelligence and natural language processing (NLP) have enabled computer systems to interpret and process queries expressed in natural language. One of the most impactful applications of this progress is Text-to-SQL, which translates natural language questions into executable SQL queries, thus allowing non-technical users to access relational databases without prior SQL expertise [1]. This democratization of data access is particularly valuable in fields such as business intelligence, data science, and decision support systems, where timely and accurate access to data is crucial [2].

As relational databases grow more complex, involving multiple tables, diverse schemas, and intricate relationships, Text-to-SQL systems provide a vital bridge between human users and structured data. Early approaches were rule-based and template-driven, gradually giving way to deep learning models. Today, large language models (LLMs) have transformed the field by enabling more flexible and accurate query generation. Their capacity for reasoning, context understanding, and code generation has improved system performance across benchmarks such as Spider, WikiSQL, and CoSQL [1], [3]. Nevertheless, challenges persist, including robustness in real-world scenarios, handling nested queries, optimizing joins, and adapting to domain-specific requirements [4].

Within this project, a two-person remote team aims to design a Text-to-SQL system that can generate accurate and optimized queries on multi-table databases using either open-source LLMs or API-based solutions. The model will be evaluated on well-established datasets, with a focus on how performance varies across different query complexity levels. Additionally, Agile methodology will be adopted to facilitate efficient project management and collaboration in a distributed setting [1].

Research Questions

1. Can the selected LLM model accurately translate natural language inputs into SQL queries on standard Text-to-SQL datasets?
2. How does the model perform across different query complexity levels (e.g., simple SELECT, JOIN, GROUP BY)?

In the continuation of this study, the second section will present a literature review of existing approaches in the field of Text-to-SQL. The third section will describe the system design and methodology employed in the project. The fourth section will include the results of model testing and evaluation, followed by a discussion of findings. Finally, the last section will conclude with reflections on the project outcomes and future directions.

References

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